Tworit Kumar Dash

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Delft, The Netherlands

Model-Based Inference Researcher

Statistical signal processing · inverse problems · spectral estimation · electromagnetics

Research Profile

I combine physical models, statistical inference, and information theory to determine what measurements contain, what they lose, and why. My work emphasizes interpretable forward models, identifiability, uncertainty, and performance bounds. I develop these ideas across Doppler spectra, phased-array sensing, precipitation retrieval, automotive radar, and electromagnetic field and antenna-array measurements.

Model-based inversion | Likelihood methods | Cramer–Rao bounds | Sampling and aliasing | Array processing | Spherical harmonics

Appointments

2025–present — **Postdoctoral Researcher** — TU Delft

Model-based array and spectral inference for phased-array atmospheric radar and antenna measurements in the Microwave Sensing, Signals and Systems group.

2020–2025 — **Doctoral Researcher** — TU Delft

Theory and estimators for short-record Doppler processing: finite-sample spectrum models, counter-aliasing, incoherent processing, and uncertainty analysis.

2016–2018 — **Calibration Engineer** — Bosch, Bangalore

Statistical modelling and threshold design for passenger-vehicle misfire and catalyst diagnostics.

Education

2025 — **PhD, Electrical Engineering** — TU Delft

On Doppler Processing for Fast Scanning Weather Radars.

2020 — **MSc, Electrical Engineering, cum laude** — TU Delft / ASTRON

Telecommunication and Sensing Systems. Thesis on computationally efficient conical horn design.

2016 — **BTech, Electronics and Instrumentation** — CET Bhubaneswar

CGPA 8.94/10; alumni merit award for best final-year undergraduate student.

Selected Publications

- **T. Dash, S. Yuan, J. Heylen, and A. Yarovoy**. Doppler Moment Estimation for Precipitation With a Phased Array Radar Using Latent Beamforming. *IEEE TGRS, published 2025. DOI: 10.1109/TGRS.2025.3645429.*
- **A. Pappas, T. Dash, et al**. Adaptive Radar Approaches for Doppler Moment Estimation. *IEEE GRSL, published 2025. DOI: 10.1109/LGRS.2025.3646459.*
- **S. Yuan, T. K. Dash, et al**. A Doppler De-Aliasing Technique for FMCW-TDM MIMO Automotive Radar. *IEEE TIM, 2025. DOI: 10.1109/TIM.2025.3623765.*
- **T. Dash, N. B. Onat, Y. Aslan, and A. Yarovoy**. Using Spherical Harmonics to Model Mutual Coupling Effects on Embedded Element Patterns. *ICEAA, 2025. DOI: 10.1109/ICEAA65662.2025.11305766.*
- **T. Dash, H. Driessen, O. A. Krasnov, and A. G. Yarovoy**. Counter-Aliasing Is Better Than De-Aliasing. *IEEE TGRS, 2024. DOI: 10.1109/TGRS.2024.3438567.*
- **T. Dash, H. Driessen, O. Krasnov, and A. Yarovoy**. Doppler Spectrum Parameter Estimation Using a Parametric Semi-Analytical Model. *IEEE TGRS, 2024. DOI: 10.1109/TGRS.2023.3338233.*

Additional Research Outputs

- "Joint Retrieval of Radial Wind, Terminal Fall Velocity, and Median Diameter From Single-Polarization Fast-Scanning Weather Radar," arXiv:2608.02364, 2026.
- "Parametric Estimation of Elevation-Doppler Profiles with Phased Array Radar for Precipitation," EuRAD, 2025.
- "Assessing Polarimetric Weather Radar Data Quality Under Cross-Polar Coupling and Sampling Limits," IEEE Radar Conference, 2025.
- "Incoherent Doppler Processing for Doppler Moment and Noise Estimation for Precipitation," URSI AT-RASC, 2024.
- "Radiation from the Open-Ended Over-Moded Cylindrical Waveguide," URSI AT-RASC, 2024.
- "Precipitation Doppler Spectrum Reconstruction With Gaussian Process Prior," IEEE CAMA, 2023.
- "Beamforming for a Fast Scanning Phased Array Weather Radar," EuRAD, 2023.

Complete list: tworit.github.io/#publications

Open Research Software and Data

TWORIT — **Theoretical waveguide solver** — 4TU.ResearchData
S-parameters and far fields of cascaded cylindrical and irregular tapered-cone waveguide structures.

WiDSE — **Wind and DSD Estimator** — 4TU.ResearchData
Model-based wind and raindrop-size retrieval from weather-radar Doppler spectra.

PSE — **Parametric Spectrum Estimator** — 4TU.ResearchData
Finite-record estimation of precipitation Doppler moments with processed radar data.

Methods and Technical Skills

Inference: Maximum likelihood, Bayesian inference, Hamiltonian Monte Carlo, Fisher information, Cramer–Rao bounds, uncertainty quantification

Signal processing: Spectral estimation, irregular sampling, aliasing analysis, Doppler processing, array processing, Gaussian processes

Electromagnetics: Waveguide theory, modal methods, antenna arrays, embedded element patterns, vector spherical harmonics

Programming: MATLAB, Python, Ruby; reproducible numerical simulation and data analysis

Tools: CST Studio Suite, FEKO, Git, Linux, Windows, macOS

Awards and Communication

2020 — **MSc graduated cum laude** — TU Delft
Telecommunication and Sensing Systems; thesis research conducted with ASTRON.

2016 — **Alumni Merit Award** — CET Bhubaneswar
Best final-year undergraduate student.

Technical speaking
Internet of Things and Ruby talks at Eurucamp 2014, Euruko 2015, and Deccan Ruby Conference.

Languages

Odia (native), English (fluent), Hindi (professional), Sanskrit (beginner).

Last updated August 2026. References available on request.